

Engineering Electromagnetic Fields

Experiment 3

Magnetic Field Distribution of Two Circular Current Loops

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1 Experiment Task Description

This experiment uses MATLAB to compute and visualise the magnetic field generated by two parallel circular current loops. The calculation is based on the Biot–Savart law:

$$d\mathbf{H} = \frac{I}{4\pi} \frac{d\mathbf{L} \times \mathbf{R}}{R^3}, \quad (1)$$

where $\mathbf{H}$ is the magnetic field intensity, $I d\mathbf{L}$ is the current element vector, and $\mathbf{R}$ is the vector from the current element to the field point.

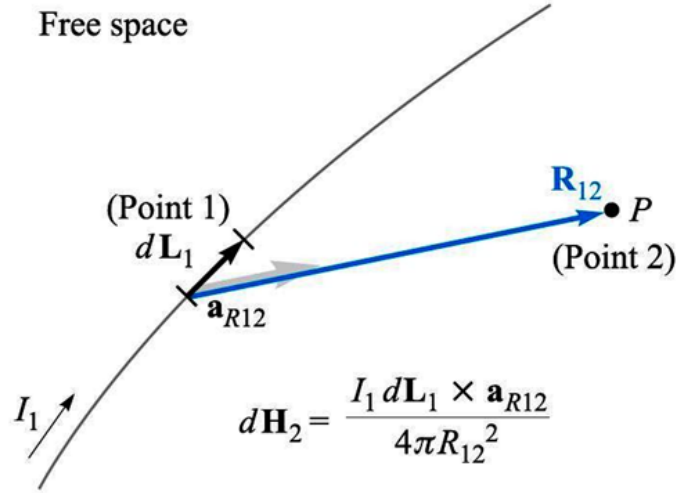


图 1: Illustration of the Biot–Savart law: magnetic field $d\mathbf{H}$ produced by a current element $I d\mathbf{L}$ at a field point P . (Source: experiment handout.)

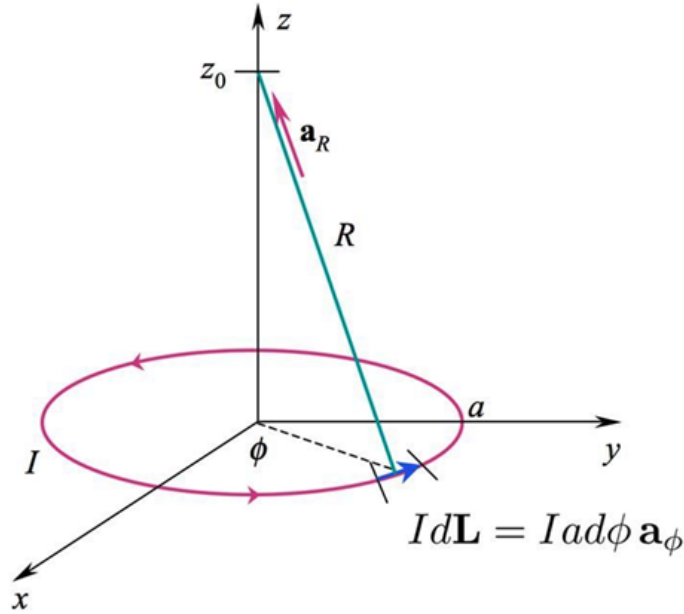


图 2: Geometry of a circular current loop and its magnetic field on the yz -plane. (Source: experiment handout.)

Fig. 1 illustrates the Biot–Savart law for a single current element. Fig. 2 shows the geometry of a circular loop in the xy -plane. For a circular loop of radius a with its centre at the origin, the magnetic field on the central axis can be derived analytically. However, for field points off the axis, a closed-form solution is difficult. We therefore adopt a numerical approach: discretise the current loop into many small current elements, compute each element’s contribution using Eq. (1), and sum them using the superposition principle.

1.1 Experiment Configuration

Two circular loops, each of radius $a = 2\text{ m}$ and carrying current $I = 500\text{ A}$, are placed parallel to the xy -plane with centres at $(0, 0, -1)$ and $(0, 0, 1)$. The magnetic field is evaluated on the yz -plane ($x = 0$). Two cases are studied:

- (i) **Case 1 — Same current direction:** Both loops carry current in the same sense. When the loop separation equals the radius ($d = a = 2\text{ m}$), this is known as a Helmholtz coil. Its key property is a nearly uniform magnetic field in the central region between the two loops.
- (ii) **Case 2 — Opposite current directions:** The upper loop's current direction is reversed. The superposition of opposing fields produces a qualitatively different pattern.

For each case, the following plots are generated: (a) 3D mesh of $|\mathbf{H}|$ over the full window $y, z \in [-5, 5]$; (b) 3D mesh zoomed to the region of interest $y \in [-2, 2], z \in [-1, 1]$; (c) contour map; (d) field-line (streamline) plot; (e) normalised vector plot. All figure titles include the student name and ID (HE Yanning, 12311512).

2 Numerical Method

The computation proceeds as follows. Each circular loop is divided into $N = 200$ straight segments of equal angular increment $\Delta\theta = 2\pi/N$. For each segment i , its mid-point coordinates $(x_{c,i}, y_{c,i})$ and vector length components $(\Delta L_{x,i}, \Delta L_{y,i})$ are computed. The field is evaluated on a regular grid of 100×100 points covering $y \in [-5, 5], z \in [-5, 5]$.

At each grid point, the contributions from all $2 \times N$ current elements are summed:

$$\mathbf{H}(0, y_j, z_k) \approx \frac{I}{4\pi} \sum_{\text{loop}=1}^2 \sum_{i=1}^N \frac{\Delta \mathbf{L}_i^{(\text{loop})} \times (\mathbf{r}_{jk} - \mathbf{r}_{c,i}^{(\text{loop})})}{|\mathbf{r}_{jk} - \mathbf{r}_{c,i}^{(\text{loop})}|^3}. \quad (2)$$

Because the observation plane is $x = 0$, the H_x component is zero by symmetry. The total magnitude is $|\mathbf{H}| = \sqrt{H_y^2 + H_z^2}$.

The complete MATLAB code is consolidated in `lab3_code.mlx` (see Appendix A), with extensive comments explaining each step.

3 Case 1: Same Current Direction (Helmholtz Coil)

In this configuration, both loops carry current in the same direction. The magnetic fields of the two loops reinforce each other in the central region.

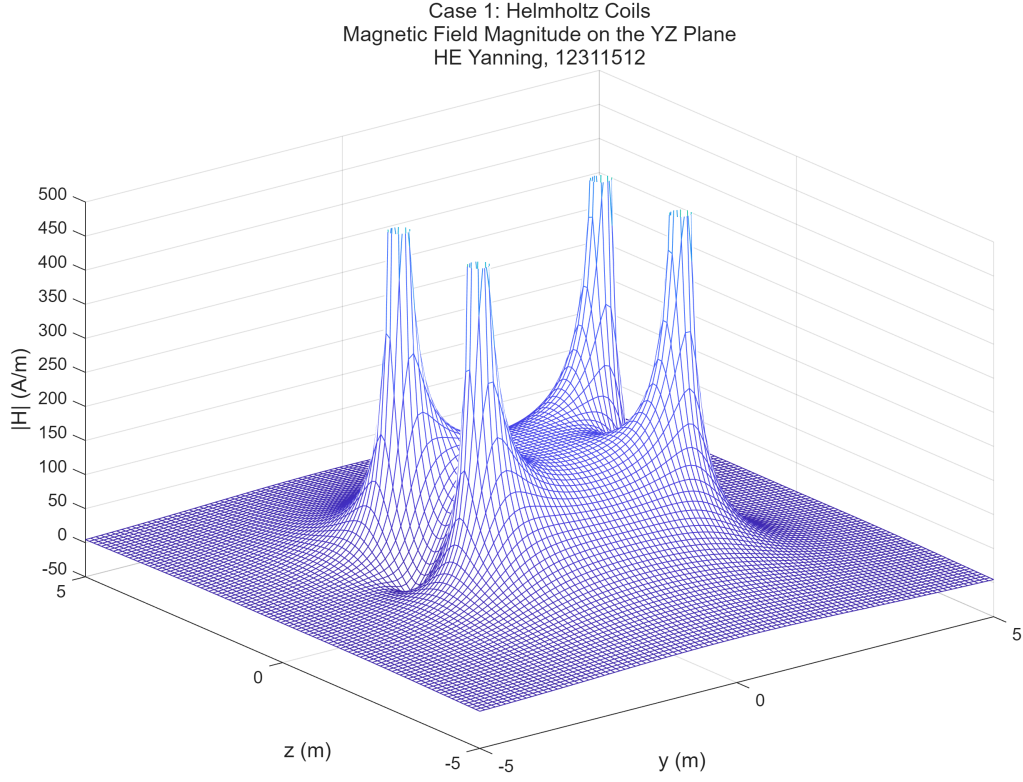


图 3: Case 1: $|\mathbf{H}|$ distribution on the yz -plane, full view ($y, z \in [-5, 5]$).

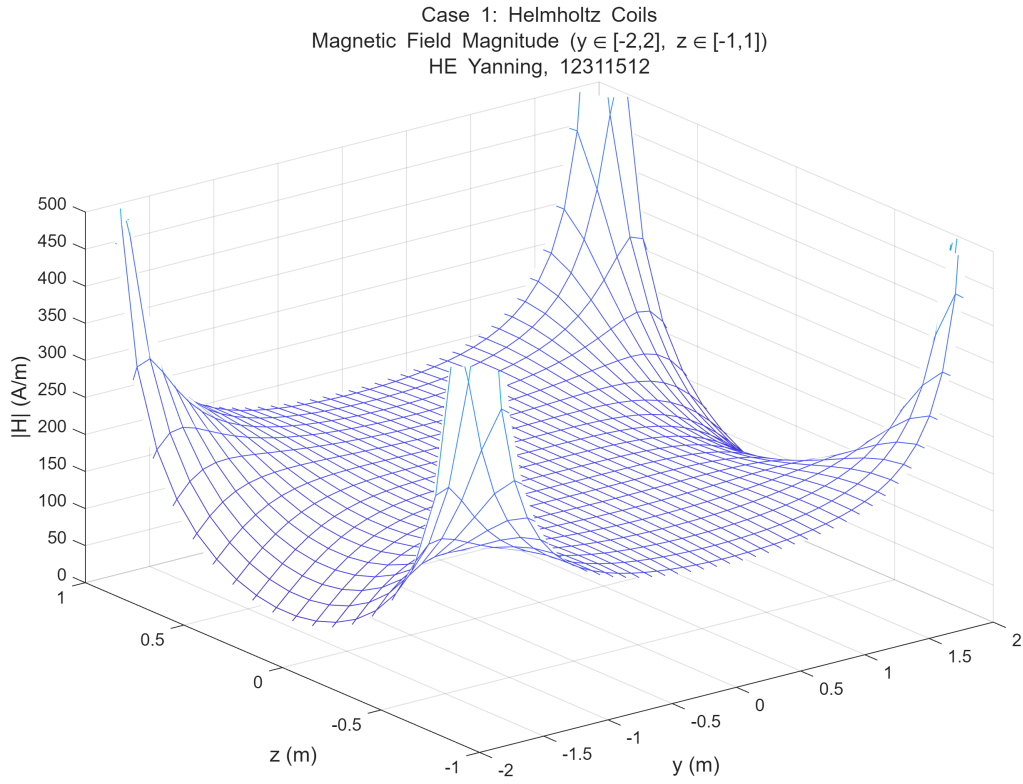


图 4: Case 1: $|\mathbf{H}|$ distribution in the region of interest ($y \in [-2, 2], z \in [-1, 1]$).

Figs. 3 and 4 show the field magnitude. Two peaks appear near $z = \pm 1$, corre-

sponding to the positions of the current-carrying loops. Inside the red rectangular region ($y \in [-2, 2], z \in [-1, 1]$), the surface is relatively flat, indicating that the field magnitude does not change dramatically.

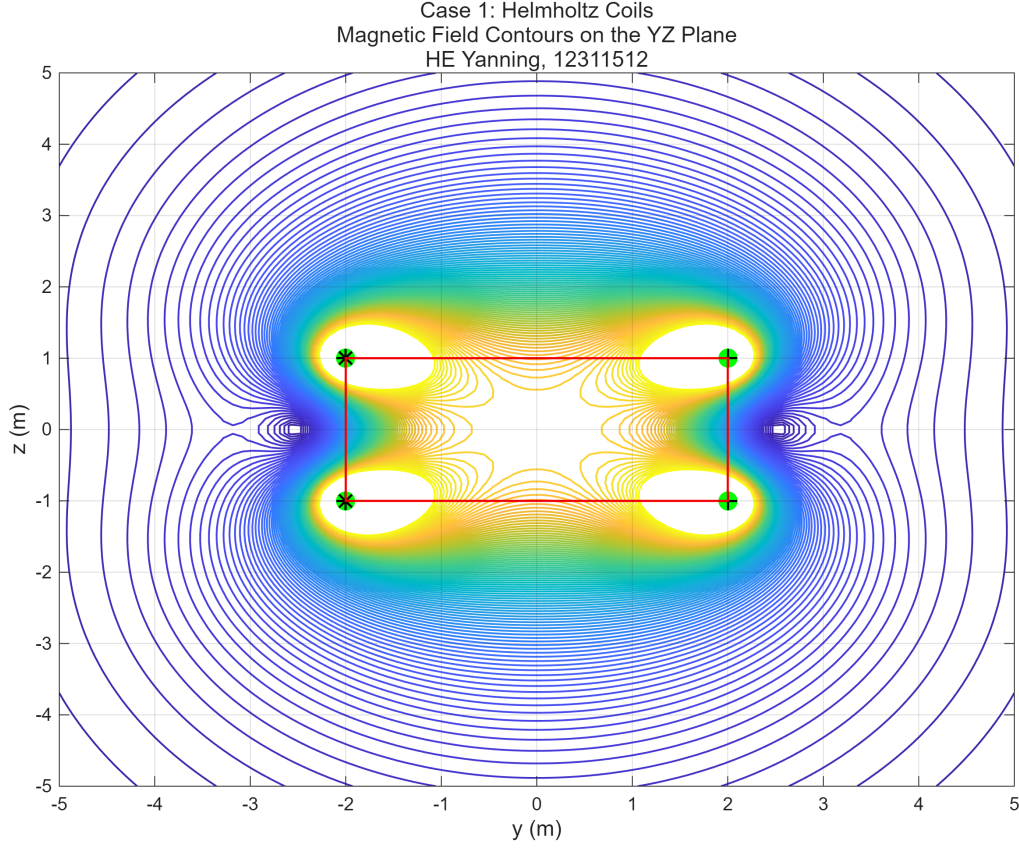


图 5: Case 1: Contour map of $|\mathbf{H}|$. The red rectangle marks the region of interest.

Fig. 5 displays isomagnetic (equal-field) contours. Green circles mark the four points where the two loops intersect the yz -plane; the overlaid markers ($*$ and $+$) indicate current direction (out of and into the page, respectively). The contours are dense near the loops (rapid field variation) and sparse near the centre (slow variation), which confirms the field uniformity.

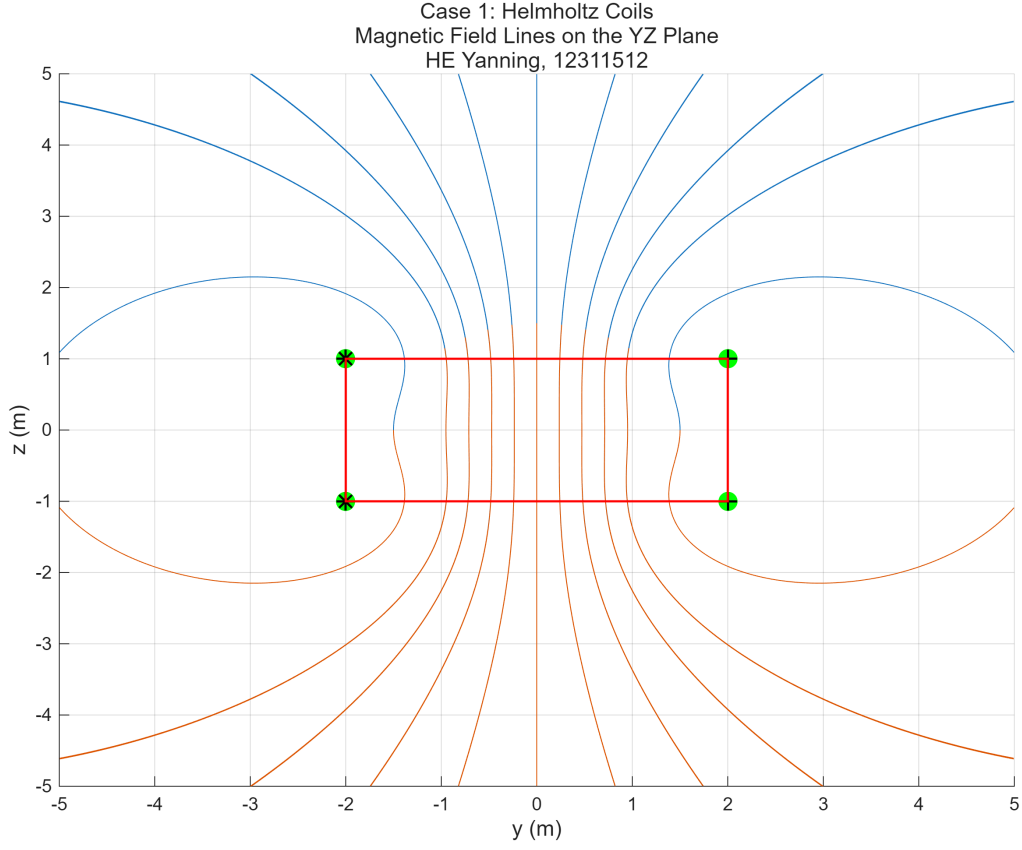


图 6: Case 1: Magnetic field lines on the yz -plane.

The streamline plot (Fig. 6) displays the field-line topology. Lines emerge from the interior of each loop, pass through the central region roughly parallel to the z -axis, and return through the exterior. The pattern is symmetric about $z = 0$.

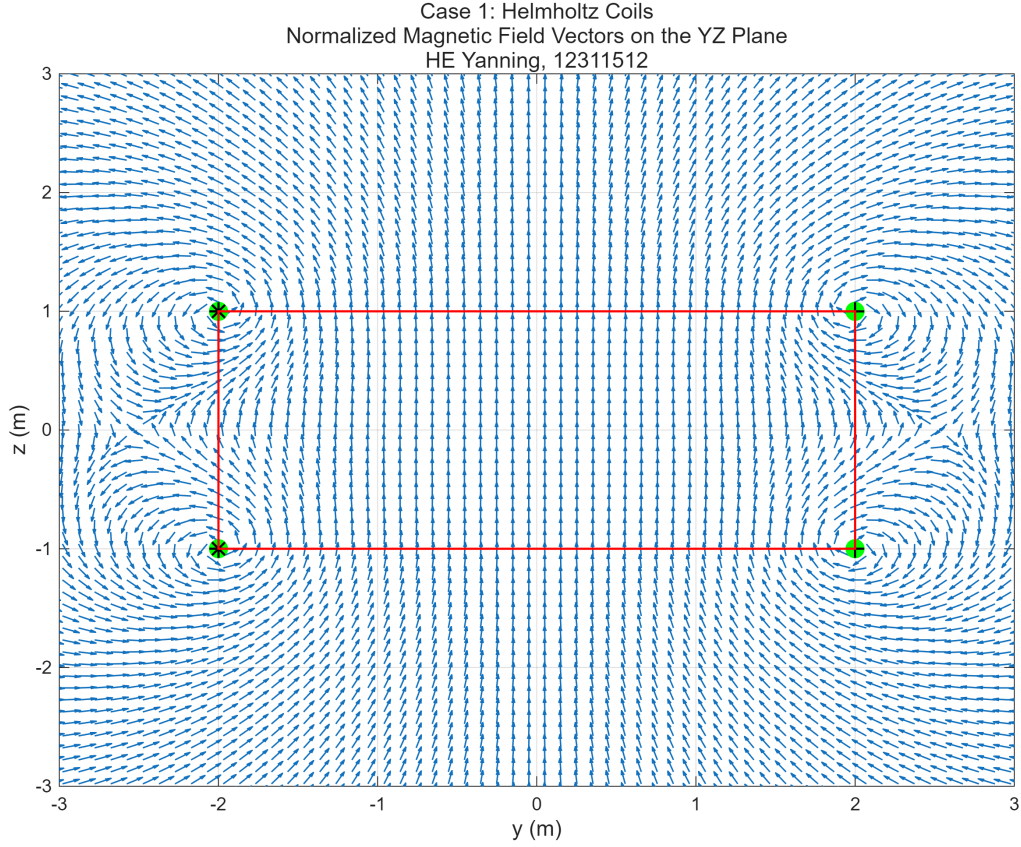


图 7: Case 1: Normalised magnetic field vectors. Arrow lengths are uniform; arrow direction shows the local field direction.

The vector plot (Fig. 7) shows the local field direction at each grid point. Inside the red rectangle, the vectors are predominantly vertical and point in the same direction, consistent with the expected behaviour of a Helmholtz coil.

Summary of Case 1: The Helmholtz-coil arrangement produces a strong, approximately uniform magnetic field in the space between the two loops. Both the mesh plot and the contour map confirm the slow spatial variation of $|\mathbf{H}|$ in the central region.

4 Case 2: Opposite Current Directions

In this configuration, the upper loop's current direction is reversed. The two fields now oppose each other in the central region.

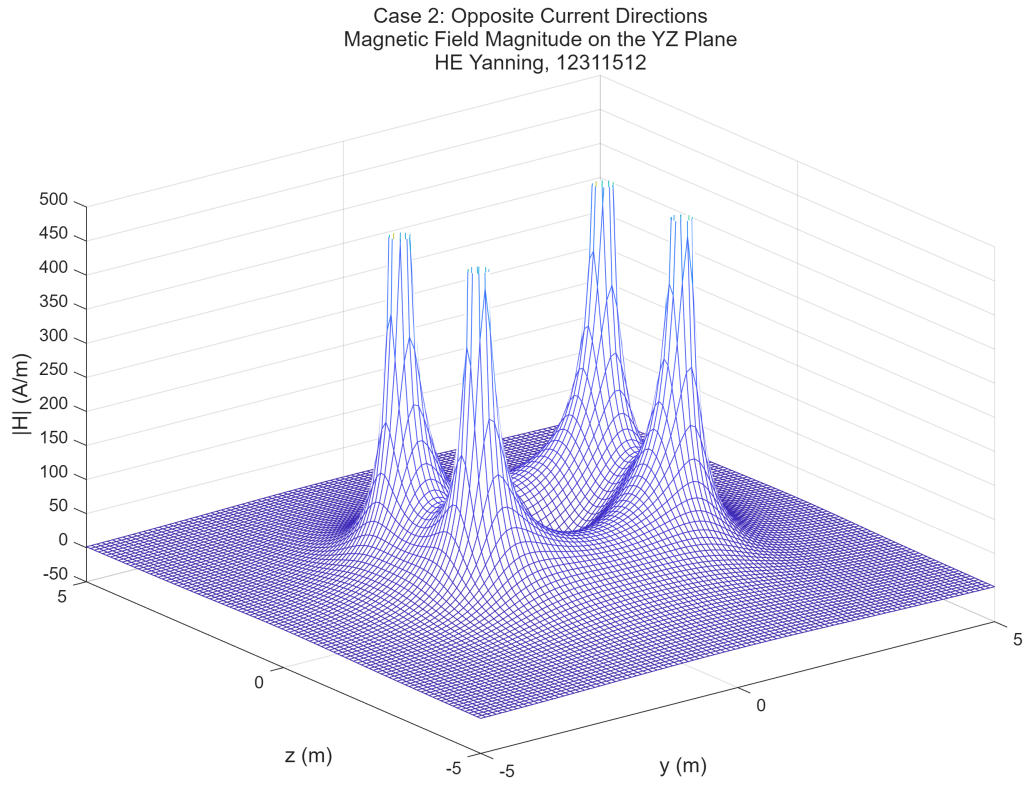


图 8: Case 2: $|\mathbf{H}|$ distribution on the yz -plane, full view.

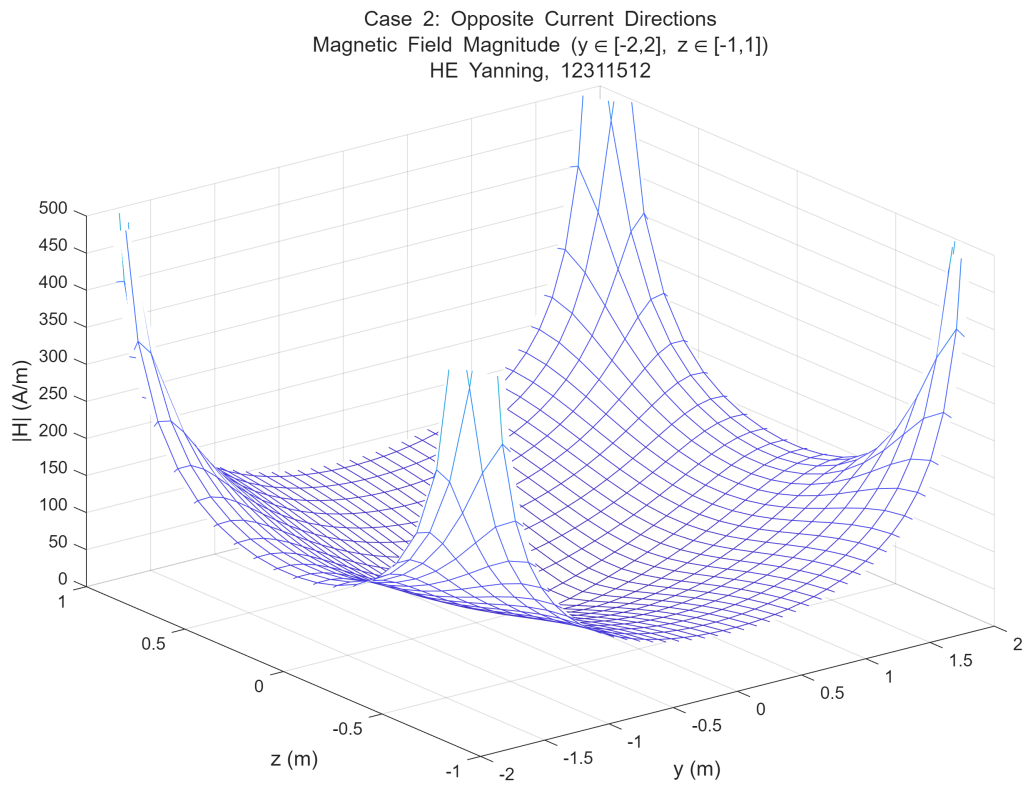


图 9: Case 2: $|\mathbf{H}|$ distribution in the region of interest. Note the weaker field near $z = 0$ compared with Case 1.

Comparing Figs. 8–9 with their Case 1 counterparts, the field in the central area is noticeably weaker. This is because the z -components from the two loops point in opposite directions near $z = 0$ and partially cancel.

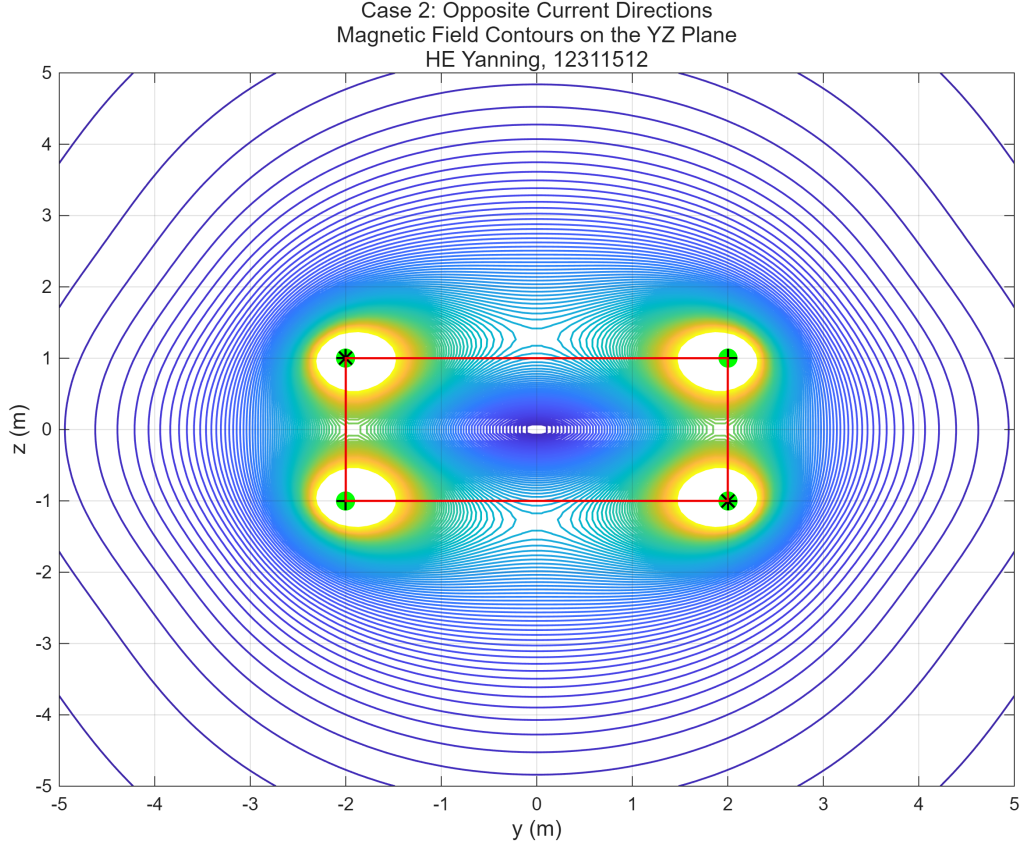


图 10: Case 2: Contour map of $|\mathbf{H}|$. Note the different marker pattern ($*$ and $+$) compared with Case 1, reflecting the reversed current in the upper loop.

The contour map (Fig. 10) shows a different marker arrangement from Case 1, reflecting the reversed current direction. The contours near the centre are more widely spaced and pulled outward, indicating a weaker and less uniform field.

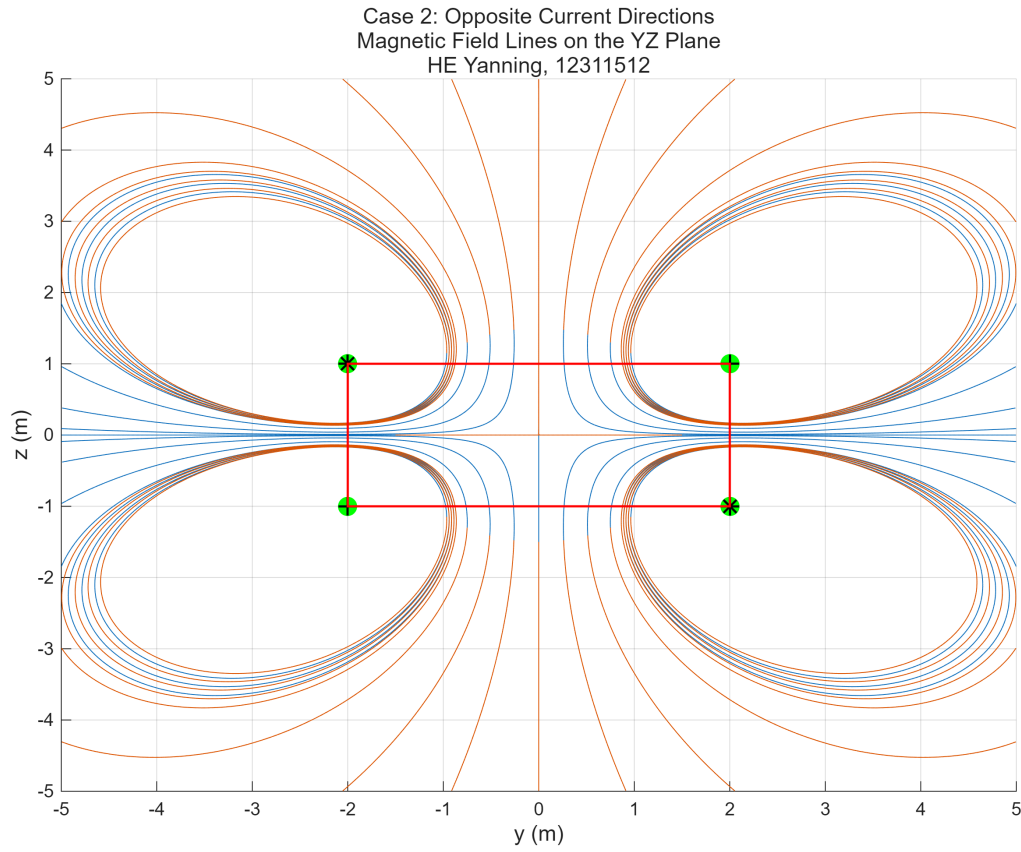


图 11: Case 2: Magnetic field lines on the yz -plane. The pattern is more complex than Case 1.

The streamline plot (Fig. 11) reveals a more complex field topology. Instead of passing straight through the central region, field lines form closed loops that directly connect the upper and lower coils.

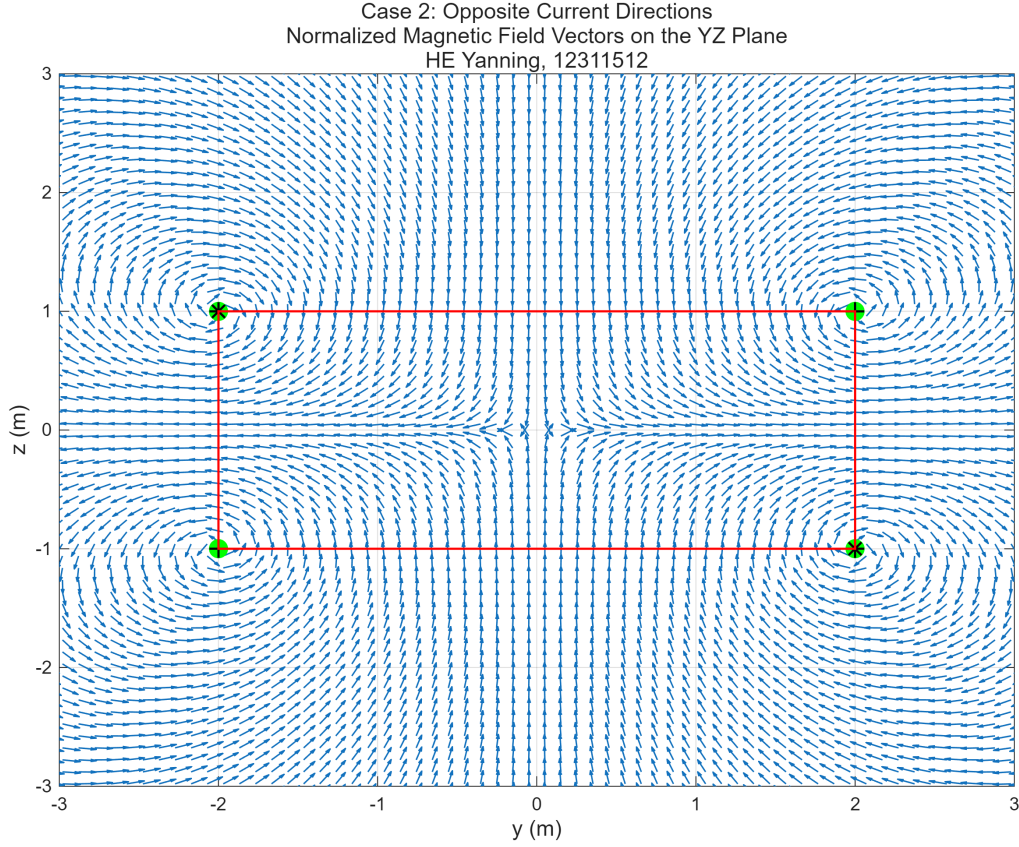


图 12: Case 2: Normalised magnetic field vectors. Note the significant horizontal components near the centre.

The vector plot (Fig. 12) confirms that the field direction in the central region is no longer predominantly vertical. Substantial horizontal (y) components appear, and the vector orientation varies considerably from point to point.

Summary of Case 2: The opposite-current configuration destroys the field uniformity. The central field is weaker and exhibits significant directional variation. This highlights the importance of current direction in multi-coil electromagnetic systems.

5 Discussion and Conclusion

5.1 Physical Interpretation

The contrasting behaviours can be understood from the superposition principle. When currents flow in the same direction (Case 1), the z -components from both loops add constructively, producing a strong axial field. The y -components are anti-symmetric about $z = 0$ and cancel at the mid-plane, leaving a clean, vertical field.

When the currents oppose each other (Case 2), the z -components cancel near the mid-plane, and the residual field is dominated by y -components that form the character-

istic inter-loop connection pattern visible in the streamline plot.

5.2 Helmholtz Coil

A Helmholtz coil is a pair of identical coaxial circular coils separated by a distance equal to their radius ($d = a$). In this experiment, $a = 2$ m and the loop separation is also 2 m, satisfying the Helmholtz condition. The numerical results confirm that Case 1 produces a nearly uniform field in the central region, which is why Helmholtz coils are widely used in laboratory applications requiring a controllable, uniform magnetic field.

5.3 Numerical Considerations

Each loop is divided into 200 segments, giving an angular resolution of 1.8° . The 100×100 grid provides a spatial resolution of 0.1 m in both y and z . These parameters produce smooth field plots while keeping the computation manageable. The total number of Biot–Savart evaluations is approximately 4×10^6 , which MATLAB handles efficiently through vectorised array operations.

5.4 Conclusion

This experiment demonstrated:

- (i) The Biot–Savart law can be implemented numerically by discretising a current loop and applying the superposition principle;
- (ii) The same-direction (Helmholtz) configuration produces a strong, uniform magnetic field between the coils;
- (iii) Reversing one coil’s current direction fundamentally alters the field distribution, destroying the central uniformity and creating a complex vector pattern;
- (iv) MATLAB provides a convenient platform for electromagnetic field computation and visualisation.

A MATLAB Source Code: lab3_code.mlx

The complete MATLAB code (including all Chinese annotations) is provided as lab3_code.mlx. The code is also available as a plain-text lab3_code.m file. Both files are located in the `source/` directory.

Code Structure

- **Parameter setup** (lines 1–30): Define loop radius, current, grid size, segment count, and the student tag.
- **Grid and loop discretisation** (lines 32–55): Build the 100×100 mesh and divide each loop into 200 current elements.
- **Case 1** (lines 57–170): Compute and plot all five figures for the Helmholtz-coil configuration (same current direction).
- **Case 2** (lines 172–290): Compute and plot all five figures for the opposite-current configuration.

Listing (plain-text version)

```

1 % =====
2 % 《电磁场理论》实验三：两个电流环在 YZ 平面上的磁场分布
3 % 报告人：HE Yanning, 12311512
4 % 南方科技大学 电子与电气工程系
5 % =====
6 % 实验说明：
7 % 两个半径 a=2 m 的圆形电流环，分别位于 z=-1 和 z=1 处（圆心在 z 轴上），
8 % 平行于 xy 平面放置，载有电流 I=500 A。在 YZ 平面（x=0）上计算磁场分布。
9 % 情形一：两个电流环电流方向相同（亥姆霍兹线圈结构）
10 % 情形二：两个电流环电流方向相反
11 % =====
12
13 clc; clear; close all;
14
15 %% ===== 参数设置 =====
16 a = 2; % 电流环半径，单位：m
17 I = 500; % 电流环中流通的电流，单位：A
18 C = I / (4 * pi); % 归并常数（毕奥-萨伐尔定律中的常数因子）
19 N = 200; % 将每个电流环等分成 N 段（分段数越大，计算越精确）

```

```

20 ym = 5; % 场域中 y 方向的范围: [-ym, ym]
21 zm = 5; % 场域中 z 方向的范围: [-zm, zm]
22 n = 100; % 将 y 轴和 z 轴各等分成 n 等份 (网格分辨率)
23
24 studentTag = 'HE Yanning, 12311512'; % 姓名及学号, 用于图片标题
25
26 %% ===== 建立网格坐标 =====
27 y = linspace(-ym, ym, n); % 将 y 轴等分成 n 等份
28 z = linspace(-zm, zm, n); % 将 z 轴等分成 n 等份
29 [Y, Z] = meshgrid(y, z); % 生成 YZ 平面上的网格矩阵
30
31 %% ===== 将电流环离散化 =====
32 % 将每个电流环的圆周角 [0, 2*pi] 等分为 N+1 个点, 取相邻两点间的线段作为电流元
33 theta0 = linspace(0, 2*pi, N+1); % 将圆周角等分为 N+1 份
34 theta1 = theta0(1:N); % 第 i 段电流元的起始角度
35 x1 = a * cos(theta1); % 第 i 段电流元的起始 x 坐标
36 y1 = a * sin(theta1); % 第 i 段电流元的起始 y 坐标
37 theta2 = theta0(2:N+1); % 第 i 段电流元的终止角度
38 x2 = a * cos(theta2); % 第 i 段电流元的终止 x 坐标
39 y2 = a * sin(theta2); % 第 i 段电流元的终止 y 坐标
40
41 xc = (x1 + x2) / 2; % 第 i 段电流元中点的 x 坐标 (用于近似计算径矢)
42 yc = (y1 + y2) / 2; % 第 i 段电流元中点的 y 坐标
43
44 dlx = x2 - x1; % 第 i 段电流元矢量 dl 的 x 分量
45 dly = y2 - y1; % 第 i 段电流元矢量 dl 的 y 分量
46 dlz = 0; % 第 i 段电流元矢量 dl 的 z 分量 (电流环在 xy 平面上, 故 z 分量为 0)
47
48 % 两个电流环的圆心 z 坐标
49 zc1 = -1; % 下方电流环圆心位于 z=-1
50 zc2 = 1; % 上方电流环圆心位于 z=1
51
52 %% =====
53 %% 情形一: 两个电流环电流方向相同
54 %% (亥姆霍兹线圈结构)
55 %% =====
56 fprintf('正在计算情形一 (电流同向, 亥姆霍兹线圈) ... \n');
57
58 % 初始化磁场分量矩阵 (全部置零)

```



```

59 Hy1 = zeros(n); % 磁场强度 H 的 y 分量
60 Hz1 = zeros(n); % 磁场强度 H 的 z 分量
61 H1 = zeros(n); % 磁场强度 H 的大小 |H| = sqrt(Hy^2 + Hz^2)
62
63 % ----- 利用毕奥-萨伐尔定律循环计算各网点上的 H 值 -----
64 % 原理：将两个电流环各分成 N 段电流元，对每个网格点，累加所有电流元
65 % 在该点产生的磁场贡献。电流元 dl 在空间某点产生的磁场为：
66 %  $dH = (I/(4\pi)) * (dl \times R) / R^3$ 
67 % 其中 R 为从电流元指向场点的矢量， $R = |R|$ 。
68 for i = 1:N
69     % 观测面为 x=0 平面，所有场点的 x 坐标均为 0
70     X = zeros(n); % 场点在 x 方向上的坐标（全为零）
71
72     rx = X - xc(i); % 径矢 R 的 x 分量：场点 x - 电流元中点 x
73     ry = Y - yc(i); % 径矢 R 的 y 分量：场点 y - 电流元中点 y
74     rz1 = Z - zc1; % 径矢 R 的 z 分量（下方电流环）：场点 z - zc1
75     rz2 = Z - zc2; % 径矢 R 的 z 分量（上方电流环）：场点 z - zc2
76
77     % 计算  $R^3 = (rx^2 + ry^2 + rz^2)^{(3/2)}$ 
78     r31 = sqrt(rx.^2 + ry.^2 + rz1.^2).^3; % 下方电流环的  $R^3$ 
79     r32 = sqrt(rx.^2 + ry.^2 + rz2.^2).^3; % 上方电流环的  $R^3$ 
80
81     % 计算叉乘  $dl \times R$  的各分量
82     % 由于  $dl = (dlx, dly, 0)$ ,  $R = (rx, ry, rz)$ :
83     %  $(dl \times R)_x = dly*rz - dlz*ry = dly*rz$  （不为零，但观测面在 x=0 上）
84     %  $(dl \times R)_y = dlz*rx - dlx*rz = -dlx*rz$  （y 分量）
85     %  $(dl \times R)_z = dlx*ry - dly*rx$  （z 分量）
86     % 情形一中两个环电流方向相同，dl 方向相同
87     dlXr_y1 = -dlx(i) .* rz1; % 下方电流环的  $(dl \times R)_y$ 
88     dlXr_y2 = -dlx(i) .* rz2; % 上方电流环的  $(dl \times R)_y$ 
89     dlXr_z = dlx(i) .* ry - dly(i) .* rx; %  $(dl \times R)_z$ （上、下环相同）
90
91     % 累加各电流元的磁场贡献： $H = H + C * (dl \times R) / R^3$ 
92     Hy1 = Hy1 + C .* dlXr_y1 ./ r31; % 累加下方环对 Hy 的贡献
93     Hy1 = Hy1 + C .* dlXr_y2 ./ r32; % 累加上方环对 Hy 的贡献
94     Hz1 = Hz1 + C .* dlXr_z ./ r31; % 累加下方环对 Hz 的贡献
95     Hz1 = Hz1 + C .* dlXr_z ./ r32; % 累加上方环对 Hz 的贡献
96     H1 = (Hy1.^2 + Hz1.^2).^0.5; % 计算合磁场大小 |H|
97 end

```

```

98
99 % ----- 图1-1: 全局磁场分布 (三维网格图) -----
100 figure('Color', 'w', 'Position', [100, 100, 900, 650]);
101 mesh(Y, Z, H1); % 绘制三维网格图
102 hold on;
103 title(sprintf(['Magnetic Field Distribution of\n' ...
104     'Two Energized Current Loops in the YZ Plane\n' ...
105     '%s'], studentTag), 'FontSize', 12); % 标题 (含姓名学号)
106 axis([-5, 5, -5, 5, -50, 500]); % 设置坐标轴范围: [y_min, y_max, z_min, z_max,
    H_min, H_max]
107 xlabel('Y axis (unit:m)', 'FontSize', 12); % y 轴标签
108 ylabel('Z axis (unit:m)', 'FontSize', 12); % z 轴标签
109 zlabel('H (unit:A/m)', 'FontSize', 12); % H 轴标签 (磁场强度大小)
110 saveas(gcf, 'case1_01_full_mesh.png'); % 保存图片
111
112 % ----- 图1-2: 局部磁场分布 (感兴趣区域 y in [-2,2], z in [-1,1])
    -----
113 figure('Color', 'w', 'Position', [100, 100, 900, 650]);
114 mesh(Y, Z, H1); % 绘制三维网格图
115 hold on;
116 title(sprintf(['Interested Region of Magnetic Field Distribution of\n' ...
117     'Two Energized Current Loops in the YZ Plane\n' ...
118     '%s'], studentTag), 'FontSize', 12);
119 axis([-2, 2, -1, 1, 0, 500]); % 聚焦到感兴趣区域
120 xlabel('Y axis (unit:m)', 'FontSize', 12);
121 ylabel('Z axis (unit:m)', 'FontSize', 12);
122 zlabel('H (unit:A/m)', 'FontSize', 12);
123 saveas(gcf, 'case1_02_local_mesh.png');
124
125 % ----- 图1-3: 等磁线图 -----
126 figure('Color', 'w', 'Position', [100, 100, 900, 650]);
127 Hmin = 0; Hmax = 200; % 等磁线 H 值的最小值和最大值
128 H0 = linspace(Hmin, Hmax, 100); % 将等磁线值等分为 100 级
129 contour(Y, Z, H1, H0); % 绘制等磁线图
130 grid on; hold on;
131 title(sprintf('Isomagnetic Line on YZ Plane\n%s', studentTag), 'FontSize', 12)
    ;
132 axis([-5, 5, -5, 5]);
133 % 标注电流环剖面位置 (用绿色圆点标记)

```

```

134 plot(-2, 1, 'o', 'MarkerFaceColor', 'g', 'MarkerSize', 12); % 上方环左端
135 plot(-2, 1, '*', 'MarkerFaceColor', 'k', 'MarkerSize', 12); % 电流方向 (* 表示
    流出纸面)
136 plot(2, 1, 'o', 'MarkerFaceColor', 'g', 'MarkerSize', 12); % 上方环右端
137 plot(2, 1, '+', 'MarkerFaceColor', 'k', 'MarkerSize', 12); % 电流方向 (+ 表示流
    入纸面)
138 plot(-2, -1, 'o', 'MarkerFaceColor', 'g', 'MarkerSize', 12); % 下方环左端
139 plot(-2, -1, '*', 'MarkerFaceColor', 'k', 'MarkerSize', 12); % 电流方向
140 plot(2, -1, 'o', 'MarkerFaceColor', 'g', 'MarkerSize', 12); % 下方环右端
141 plot(2, -1, '+', 'MarkerFaceColor', 'k', 'MarkerSize', 12); % 电流方向
142 rectangle('Position', [-2, -1, 4, 2], 'EdgeColor', 'r'); % 红色矩形框标记感兴趣
    区域
143 xlabel('Y axis (unit:m)', 'FontSize', 12);
144 ylabel('Z axis (unit:m)', 'FontSize', 12);
145 saveas(gcf, 'case1_03_contour.png');
146
147 % ----- 图1-4: 磁力线分布图 -----
148 figure('Color', 'w', 'Position', [100, 100, 900, 650]);
149 % 设置磁力线起始圆的参数 (在半径为 1.5 的圆上取种子点)
150 theta = [0 50 60 70 80 90 100 110 120 130 180] .* pi / 180; % 种子点角度 (弧
    度)
151 ys = 1.5 * cos(theta); % 磁力线起始点的 y 坐标
152 zs = 1.5 * sin(theta); % 磁力线起始点的 z 坐标
153 streamline(Y, Z, Hy1, Hz1, ys, zs); % 从起始圆向外绘制磁力线
154 streamline(Y, Z, -Hy1, -Hz1, ys, zs); % 从起始圆向内绘制磁力线 (反方向)
155 title(sprintf('Magnetic Field Lines on YZ Plane\n%s', studentTag), 'FontSize',
    12);
156 hold on;
157 axis([-5, 5, -5, 5]);
158 % 标注电流环剖面位置 (同上)
159 plot(-2, 1, 'o', 'MarkerFaceColor', 'g', 'MarkerSize', 12);
160 plot(-2, 1, '*', 'MarkerFaceColor', 'k', 'MarkerSize', 12);
161 plot(2, 1, 'o', 'MarkerFaceColor', 'g', 'MarkerSize', 12);
162 plot(2, 1, '+', 'MarkerFaceColor', 'k', 'MarkerSize', 12);
163 plot(-2, -1, 'o', 'MarkerFaceColor', 'g', 'MarkerSize', 12);
164 plot(-2, -1, '*', 'MarkerFaceColor', 'k', 'MarkerSize', 12);
165 plot(2, -1, 'o', 'MarkerFaceColor', 'g', 'MarkerSize', 12);
166 plot(2, -1, '+', 'MarkerFaceColor', 'k', 'MarkerSize', 12);
167 rectangle('Position', [-2, -1, 4, 2], 'EdgeColor', 'r');

```

```

168 xlabel('Y axis (unit:m)', 'FontSize', 12);
169 ylabel('Z axis (unit:m)', 'FontSize', 12);
170 saveas(gcf, 'case1_04_streamline.png');
171
172 % ----- 图1-5: 磁场强度矢量图 -----
173 figure('Color', 'w', 'Position', [100, 100, 900, 650]);
174 Hy1_norm = Hy1 ./ H1; % 将 Hy 归一化 (只保留方向信息)
175 Hz1_norm = Hz1 ./ H1; % 将 Hz 归一化
176 quiver(Y, Z, Hy1_norm, Hz1_norm); % 绘制矢量箭头图
177 hold on;
178 title(sprintf(['Magnetic Field Intensity Vector of\n' ...
179     'Two Energized Current Loops in the YZ Plane\n' ...
180     '%s'], studentTag), 'FontSize', 12);
181 axis([-3, 3, -3, 3]); % 矢量图使用较小的坐标范围以看清箭头方向
182 % 标注电流环剖面位置
183 plot(-2, 1, 'o', 'MarkerFaceColor', 'g', 'MarkerSize', 12);
184 plot(-2, 1, '*', 'MarkerFaceColor', 'k', 'MarkerSize', 12);
185 plot(2, 1, 'o', 'MarkerFaceColor', 'g', 'MarkerSize', 12);
186 plot(2, 1, '+', 'MarkerFaceColor', 'k', 'MarkerSize', 12);
187 plot(-2, -1, 'o', 'MarkerFaceColor', 'g', 'MarkerSize', 12);
188 plot(-2, -1, '*', 'MarkerFaceColor', 'k', 'MarkerSize', 12);
189 plot(2, -1, 'o', 'MarkerFaceColor', 'g', 'MarkerSize', 12);
190 plot(2, -1, '+', 'MarkerFaceColor', 'k', 'MarkerSize', 12);
191 rectangle('Position', [-2, -1, 4, 2], 'EdgeColor', 'r');
192 xlabel('Y axis (unit:m)', 'FontSize', 12);
193 ylabel('Z axis (unit:m)', 'FontSize', 12);
194 saveas(gcf, 'case1_05_vector.png');
195
196 fprintf('情形一的 5 张图片已生成完毕。\\n');
197
198 %% =====
199 %% 情形二: 两个电流环电流方向相反
200 %% =====
201 fprintf('正在计算情形二 (电流反向) ...\\n');
202
203 % 重新初始化磁场分量矩阵
204 Hy2 = zeros(n);
205 Hz2 = zeros(n);
206 H2 = zeros(n);

```

```

207
208 % ----- 循环计算各网点上的 H 值 -----
209 % 注意：情形二与情形一的关键区别在于上方电流环的电流方向相反，
210 % 即上方环的 dl 分量取反号。在计算 dl×R 时：
211 % 下方环的 y 分量仍为 dlXr_y1 = -dlx*rz1（与情形一相同）
212 % 上方环的 y 分量变为 dlXr_y2 = -(-dlx)*rz2 = +dlx*rz2（取反）
213 % z 分量上方环也取反
214 for i = 1:N
215     X = zeros(n);
216     rx = X - xc(i);
217     ry = Y - yc(i);
218     rz1 = Z - (-1); % 下方电流环圆心 z = -1
219     rz2 = Z - 1; % 上方电流环圆心 z = 1（此处直接用数字，避免与向量混淆）
220
221     r31 = sqrt(rx.^2 + ry.^2 + rz1.^2).^3;
222     r32 = sqrt(rx.^2 + ry.^2 + rz2.^2).^3;
223
224     % 情形二：上方环电流方向相反 → dl 取反 → 叉乘结果也取反
225     dlXr_y1 = -dlx(i) .* rz1; % 下方环：(dl×R)_y（与情形一相同）
226     dlXr_y2 = -(-dlx(i)) .* rz2; % 上方环：(dl×R)_y（取反号！）
227     dlXr_z1 = -dlx(i) .* ry + dly(i) .* rx; % 下方环：(dl×R)_z（取反！注意与情形
        一的区别）
228     dlXr_z2 = dlx(i) .* ry - dly(i) .* rx; % 上方环：(dl×R)_z（与情形一相同）
229
230     Hy2 = Hy2 + C .* dlXr_y1 ./ r31; % 累加下方环对 Hy 的贡献
231     Hy2 = Hy2 + C .* dlXr_y2 ./ r32; % 累加上方环对 Hy 的贡献（方向相反）
232     Hz2 = Hz2 + C .* dlXr_z1 ./ r31; % 累加下方环对 Hz 的贡献（方向相反）
233     Hz2 = Hz2 + C .* dlXr_z2 ./ r32; % 累加上方环对 Hz 的贡献
234     H2 = (Hy2.^2 + Hz2.^2).^0.5; % 计算合磁场大小
235 end
236
237 % ----- 图2-1：全局磁场分布（三维网格图） -----
238 figure('Color', 'w', 'Position', [100, 100, 900, 650]);
239 mesh(Y, Z, H2);
240 hold on;
241 title(sprintf(['Magnetic Field Distribution of\n' ...
242     'Two Opposite-Direction Current Loops in the YZ Plane\n' ...
243     '%s'], studentTag), 'FontSize', 12);
244 axis([-5, 5, -5, 5, -50, 500]);

```

```

245 xlabel('Y axis (unit:m)', 'FontSize', 12);
246 ylabel('Z axis (unit:m)', 'FontSize', 12);
247 zlabel('H (unit:A/m)', 'FontSize', 12);
248 saveas(gcf, 'case2_01_full_mesh.png');
249
250 % ----- 图2-2: 局部磁场分布 (感兴趣区域) -----
251 figure('Color', 'w', 'Position', [100, 100, 900, 650]);
252 mesh(Y, Z, H2);
253 hold on;
254 title(sprintf(['Interested Region of Magnetic Field Distribution of\n' ...
255     'Two Opposite-Direction Current Loops in the YZ Plane\n' ...
256     '%s'], studentTag), 'FontSize', 12);
257 axis([-2, 2, -1, 1, 0, 500]);
258 xlabel('Y axis (unit:m)', 'FontSize', 12);
259 ylabel('Z axis (unit:m)', 'FontSize', 12);
260 zlabel('H (unit:A/m)', 'FontSize', 12);
261 saveas(gcf, 'case2_02_local_mesh.png');
262
263 % ----- 图2-3: 等磁线图 -----
264 % 注意: 情形二中上方环电流反向 → 电流方向标记与情形一不同
265 % 下方环: 左侧 * (流出)、右侧 + (流入) → 与情形一相同
266 % 上方环: 左侧 + (流入)、右侧 * (流出) → 与情形一相反!
267 figure('Color', 'w', 'Position', [100, 100, 900, 650]);
268 Hmin = 0; Hmax = 200;
269 H0 = linspace(Hmin, Hmax, 100);
270 contour(Y, Z, H2, H0);
271 grid on; hold on;
272 title(sprintf('Isomagnetic Line on YZ Plane\n%s', studentTag), 'FontSize', 12)
273 ;
274 axis([-5, 5, -5, 5]);
275 plot(-2, 1, 'o', 'MarkerFaceColor', 'g', 'MarkerSize', 12); % 上方环左端
276 plot(-2, 1, '*', 'MarkerFaceColor', 'k', 'MarkerSize', 12); % 电流方向 (*流出)
277 plot(2, 1, 'o', 'MarkerFaceColor', 'g', 'MarkerSize', 12); % 上方环右端
278 plot(2, 1, '+', 'MarkerFaceColor', 'k', 'MarkerSize', 12); % 电流方向 (+流入)
279 plot(-2, -1, 'o', 'MarkerFaceColor', 'g', 'MarkerSize', 12); % 下方环左端
280 plot(-2, -1, '+', 'MarkerFaceColor', 'k', 'MarkerSize', 12); % 注意: 下方环此处
281 % 为 + (流入)
282 plot(2, -1, 'o', 'MarkerFaceColor', 'g', 'MarkerSize', 12); % 下方环右端
283 plot(2, -1, '*', 'MarkerFaceColor', 'k', 'MarkerSize', 12); % 注意: 下方环此处

```


为 * (流出)

```

282 rectangle('Position', [-2, -1, 4, 2], 'EdgeColor', 'r');
283 xlabel('Y axis (unit:m)', 'FontSize', 12);
284 ylabel('Z axis (unit:m)', 'FontSize', 12);
285 saveas(gcf, 'case2_03_contour.png');
286
287 % ----- 图2-4: 磁力线分布图 -----
288 % 注意: 情形二中磁场分布更复杂, 需要使用更多的种子点来完整绘制磁力线
289 figure('Color', 'w', 'Position', [100, 100, 900, 650]);
290 theta = [0 50 60 70 80 90 100 110 120 130 180 ...
291          230 240 250 260 270 280 290 300 310] .* pi / 180; % 增加了更多种子角度
292 ys = 1.5 * cos(theta);
293 zs = 1.5 * sin(theta);
294 streamline(Y, Z, Hy2, Hz2, ys, zs); % 向外绘制磁力线
295 streamline(Y, Z, -Hy2, -Hz2, ys, zs); % 向内绘制磁力线
296 title(sprintf('Magnetic Field Lines on YZ Plane\n%s', studentTag), 'FontSize',
        12);
297 hold on;
298 axis([-5, 5, -5, 5]);
299 plot(-2, 1, 'o', 'MarkerFaceColor', 'g', 'MarkerSize', 12);
300 plot(-2, 1, '*', 'MarkerFaceColor', 'k', 'MarkerSize', 12);
301 plot(2, 1, 'o', 'MarkerFaceColor', 'g', 'MarkerSize', 12);
302 plot(2, 1, '+', 'MarkerFaceColor', 'k', 'MarkerSize', 12);
303 plot(-2, -1, 'o', 'MarkerFaceColor', 'g', 'MarkerSize', 12);
304 plot(-2, -1, '+', 'MarkerFaceColor', 'k', 'MarkerSize', 12);
305 plot(2, -1, 'o', 'MarkerFaceColor', 'g', 'MarkerSize', 12);
306 plot(2, -1, '*', 'MarkerFaceColor', 'k', 'MarkerSize', 12);
307 rectangle('Position', [-2, -1, 4, 2], 'EdgeColor', 'r');
308 xlabel('Y axis (unit:m)', 'FontSize', 12);
309 ylabel('Z axis (unit:m)', 'FontSize', 12);
310 saveas(gcf, 'case2_04_streamline.png');
311
312 % ----- 图2-5: 磁场强度矢量图 -----
313 figure('Color', 'w', 'Position', [100, 100, 900, 650]);
314 Hy2_norm = Hy2 ./ H2; % 归一化 Hy
315 Hz2_norm = Hz2 ./ H2; % 归一化 Hz
316 quiver(Y, Z, Hy2_norm, Hz2_norm);
317 hold on;
318 title(sprintf('Magnetic Field Intensity Vector of\n' ...

```

```

319     'Two Opposite-Direction Current Loops in the YZ Plane\n' ...
320     '%s'], studentTag), 'FontSize', 12);
321 axis([-3, 3, -3, 3]);
322 plot(-2, 1, 'o', 'MarkerFaceColor', 'g', 'MarkerSize', 12);
323 plot(-2, 1, '*', 'MarkerFaceColor', 'k', 'MarkerSize', 12);
324 plot(2, 1, 'o', 'MarkerFaceColor', 'g', 'MarkerSize', 12);
325 plot(2, 1, '+', 'MarkerFaceColor', 'k', 'MarkerSize', 12);
326 plot(-2, -1, 'o', 'MarkerFaceColor', 'g', 'MarkerSize', 12);
327 plot(-2, -1, '+', 'MarkerFaceColor', 'k', 'MarkerSize', 12);
328 plot(2, -1, 'o', 'MarkerFaceColor', 'g', 'MarkerSize', 12);
329 plot(2, -1, '*', 'MarkerFaceColor', 'k', 'MarkerSize', 12);
330 rectangle('Position', [-2, -1, 4, 2], 'EdgeColor', 'r');
331 xlabel('Y axis (unit:m)', 'FontSize', 12);
332 ylabel('Z axis (unit:m)', 'FontSize', 12);
333 saveas(gcf, 'case2_05_vector.png');
334
335 fprintf('情形二的 5 张图片已生成完毕。\\n');
336 fprintf('全部 10 张图片已生成。\\n');
337
338 % =====
339 % 实验总结：
340 % 1. 情形一（亥姆霍兹线圈）：两个线圈之间的磁场分布比较均匀，
341 % 尤其在 y in [-2,2], z in [-1,1] 区域内，磁场大小和方向变化不大，
342 % 这正是亥姆霍兹线圈的重要特性。
343 % 2. 情形二（电流反向）：中心区域的磁场明显减弱，磁力线呈现复杂的
344 % 连接模式，矢量方向变化剧烈，不再具备均匀性。
345 % 3. 毕奥-萨伐尔定律的数值计算方法简单直观，通过将连续电流环离散为
346 % 有限个电流元，再利用叠加原理，可以高效地求解任意位置的磁场。
347 % =====

```